INCH POUND

MIL-C-3098/99B
5 May 1993

SUPERSEDING
MIL-C-3098/99A
2 November 1983

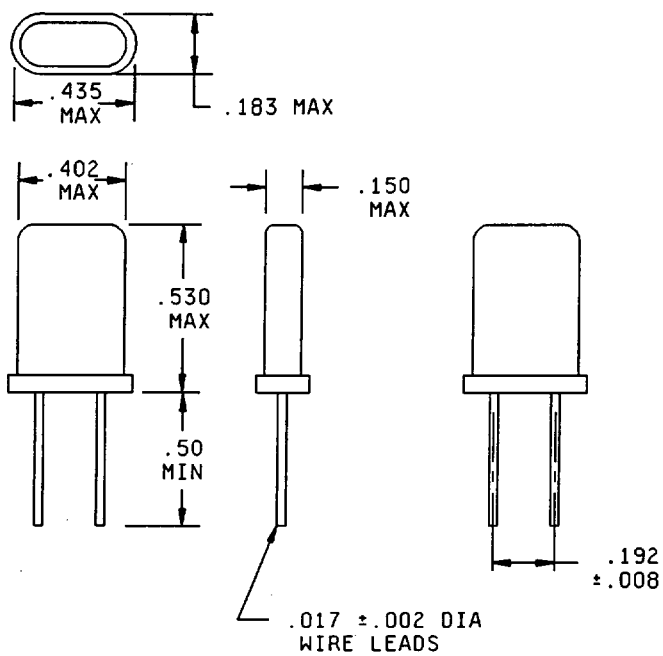
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR122/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth mechanical overtone; controlled; series resonance.



Inches	mm
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR 122/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:49Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Holder: HC-18/U (MIL-H-10056/9) or HC-43/U.

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50 to 125 MHz.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerances:

Operating temperature range: ±10 ppm.

Reference temperature: ±10 ppm.

Room temperature: ±80 ppm.

Frequency stability. ±5 ppm.

Mode of oscillation: Fifth mechanical overtone.

Rated drive level: 1.0 ±0.2 mW.

Resonance: Series.

Reference temperature: +85°C ±1°C.

Temperature ranges:

Operating (controlled): +80°C to +90°C, inclusive.

Operable: -55°C to +80°C, inclusive.

Equivalent resistance:

MHz	Ohms, max
50 to 100	50
100 to 125	60

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:49Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:

Army - ER
Navy - EC
Air Force - 85

Review activities:

Air Force - 17, 19

Preparing activity:

Army - ER

Agent:

DLA - ES

(Project 5955-0678-14)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:49Z
Check the source to verify that this is the current version before use.